

RAYANE DAKHLAOU

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Paris, France - French Citizenship

EDUCATION

PSL University – ENS Paris & Mines Paris & Paris-Dauphine

Sept. 2025 – Sept. 2026

MSc in IASD (Artificial Intelligence, Systems & Data) - Mathematics Track

Ranked 7th

- Research-oriented Master's in Applied Mathematics, Learning, and Modeling that includes courses such as:
Reinforcement Learning · Advanced Machine Learning · Advanced Statistics (High-Dimensional, Bayesian) · Deep Learning for Computer Vision · 3D Vision · LLM for Code and Proof · Kernel Methods · Optimal Transport · Representation Learning

Télécom Paris – Institut Polytechnique de Paris

Sept. 2024 – Apr. 2027

MSc in Applied Mathematics & Artificial Intelligence (Engineering Degree)

GPA: 3.94/4.00

- Artificial Intelligence & Signal Processing:** Machine Learning, Deep Learning (CNN, Transformers, GANs, VAE), Optimization, Statistics (Linear Models), Audio & Speech Processing, Time-Series Analysis.
- Applied Mathematics & Probability:** Martingales, Markov Chains, Stochastic Processes (Brownian Motion, Poisson), Numerical Analysis, Statistical Learning

Télécom Paris – Institut Polytechnique de Paris

Sept. 2021 – June 2024

BSc in Mathematics & Computer Science

GPA: 3.82/4.00

- Fundamental curriculum in Mathematics, Computer Science and Physics preparing for the Master of Science in Engineering.

RESEARCH EXPERIENCE

AI Research Intern – Siemens AG

Apr. 2026 – Sept. 2026

Dept. Foundational Technologies

Munich, Germany

- Designed *BRep-JEPA*, a novel Self-Supervised Representation Learning framework for Boundary-Representation (3D CAD) built on a Joint-Embedding Predictive Architecture (JEPA).
- Built the encoder over a geometric Attributed Adjacency Graph (gAAG), coupling Cross-Scale Mutual Attention with edge-aware Message Passing (Graph Transformer); benchmarked an enhanced Point Transformer v3 (Utonia) as an alternative encoder.
- Reached 90.11 ± 0.93 Semantic Segmentation accuracy on MFInstSeg with 0.1% of labels (44 parts), +3.2 over the strongest method published on the official splits, with a 4.8x smaller encoder.
- Panoptic Segmentation with PQ up to 92.4 with 3% of labels, the first such results reported under label scarcity.
- Isolated the prediction target with encoder, data, masking, schedule and evaluation held fixed: latent prediction gains +47.2 frozen-probe accuracy points over a paired from-scratch control (disjoint 95% confidence intervals).
- Owned the pipeline end to end: raw 3D CAD data processing, pretraining runs, and an evaluation protocol built on paired from-scratch controls, frozen-probe and few-shot transfer, and FLOP-level cost accounting.
- Master's thesis: "Self-Supervised Representation Learning on 3D CAD Models via Latent Prediction", submission to a top ML/vision venue in preparation.

PROJECTS & ACADEMIC RESEARCH

Kernel Methods for Image Classification | [GitHub](#) | [Kaggle](#)

- Built a CIFAR-10 image classifier using kernel methods (with no deep-learning library) and ranked **3rd out of 44 teams** on the Kaggle Data Challenge leaderboard.
- Designed a hierarchical feature extraction pipeline fusing Convolutional Kernel Networks (512 k-means++ filters, Spatial Pyramid Matching) and HOG descriptors, optimized via PCA reduction and an RBF-Kernel Ridge Regression.

GAN Latent Space Tuning | [GitHub](#)

- Investigated the precision-recall trade-off in GAN by training f -GANs across multiple divergences (KL, RKL, JS, BCE).
- Implemented Soft Truncation & Discriminator Rejection Sampling to improve post-hoc sample quality.
- Evaluated with FID and kNN-based precision/recall, with ablations over architecture and training hyperparameters.

DeepSDF: Implicit Neural 3D Shape Representation | [Report](#)

- Reproduced DeepSDF to learn continuous signed-distance fields as implicit neural representations of 3D models via latent-code optimization.
- Evaluated the architecture through marching-cubes reconstruction and latent interpolation, with ablations over latent width and skip connections.

TECHNICAL SKILLS

Programming: Python (Advanced) · C · CUDA · SQL · TypeScript · $\LaTeX$

ML & scientific stack: PyTorch · NumPy · Pandas · scikit-learn · TensorFlow/Keras · Numba · Git · Linux · GPU training

Languages: French (native) · English (full professional) · Arabic (intermediate) · German (beginner)

Achievements: Finalist in IBM × Qubit Pharmaceuticals × Margo Hackathon (56-hours), Finalist in IBM x TBF Data Challenge